Spatiotemporal Coordination for Grid-Responsive AI Datacenters: Unlocking Flexibility Through Multi-Region Migration

Karthik Mattu
Golisano College of Computing and Information Sciences
Rochester Institute of Technology
km5503@rit.edu

***Abstract*—Grid demand response (DR) programs require large electricity consumers to curtail load during stress events. Datacenters—increasingly dominant grid loads—typically respond with local temporal actions: throttling GPUs via Dynamic Voltage and Frequency Scaling (DVFS) or pausing jobs. These strategies impose measurable service degradation without exploiting a key structural property: regional grid stress events are temporally offset, so one region's peak rarely coincides with another's.**

We present CoordDR, a mixed-integer linear programming (MILP) coordinator that adds geographic workload migration as a control knob alongside DVFS and job pausing. Using five years (2020–2024) of hourly load data from CAISO, PJM, and BPA—three regions chosen because all major hyperscalers operate datacenters across them, connected by private fiber—we confirm the complementarity thesis: pairwise stress correlations are $\rho < 0.30$, fleet availability exceeds 95%, and simultaneous tri-regional stress occurs in fewer than 0.5% of hours.

To isolate the value of spatial flexibility, we compare CoordDR against an identical MILP with migration disabled, ensuring any performance gap is attributable purely to the spatial degree of freedom. Across 12,840 stress-event hours and four workload ensembles, both strategies achieve identical curtailment. However, CoordDR reduces quality-of-service (QoS) degradation cost by 87–96% depending on region: 95% at PJM, 93% at CAISO, and 87% at BPA. The dominant mechanism is substitution: jobs that the temporal-only strategy must pause or aggressively throttle are instead migrated at 1–2% overhead, though the exact action displaced varies with fleet composition and headroom availability.

***Index Terms*—demand response, datacenter coordination, workload migration, mixed-integer programming, grid stress complementarity**

I. Introduction

Datacenters account for 1–2% of global electricity consumption, a fraction growing rapidly with large language model training and inference workloads [1]. This growth is bottlenecked by aging grid infrastructure: capacity market prices in PJM have surged 11-fold as datacenter load strains forward procurement [2], [3]. Grid operators respond to demand peaks by issuing demand response (DR) signals that require enrolled consumers to curtail load within minutes. Compliance is contractually mandated; failure exposes operators to financial penalties.

Existing datacenter DR strategies are *temporal*: defer batch work to off-peak hours, throttle processors via Dynamic Voltage and Frequency Scaling (DVFS), or pause non-critical jobs. Emerald AI's Phoenix field demonstration [4] represents the state of the art—DVFS and job pausing coordinated across a single 256-GPU cluster, achieving approximately 25% power reduction. The fundamental limitation is that temporal-only approaches impose measurable QoS degradation: DVFS at 30% clock reduction degrades GPU throughput by $\sim$30%; pausing training jobs incurs downtime and wastes GPU-hours between the last checkpoint and suspension. Critically, latency-sensitive inference workloads ("Flex 0") cannot be throttled at all without violating SLAs, leaving a significant and growing fraction of datacenter load unaddressed [4].

We observe that datacenter operators running fleets across multiple grid regions have an alternative: *spatial* response via live workload migration. Migration removes load from the stressed region without degrading the job—it merely adds a small re-warm overhead (1–2% throughput loss [5], [6]). This is viable only if the *destination* region has available headroom, *i.e.,* its load is below its own stress threshold. The key question is whether regional stress events are temporally complementary enough to make spatial response reliable.

a) Thesis: We show that for CAISO, PJM, and BPA—three US grid regions where all major hyperscalers (AWS, Google, Microsoft) operate production datacenters connected by private fiber networks—pairwise stress correlations are $\rho < 0.30$, fleet availability exceeds 95%, and simultaneous tri-regional stress occurs in fewer than 0.5% of hours over 2020–2024. This structural complementarity, driven by fundamentally different generation portfolios (solar, thermal, hydro), enables a MILP coordinator that substitutes migration for degradation.

b) Contributions:

1) A five-year empirical analysis of grid stress complementarity across CAISO, PJM, and BPA, demonstrating structural temporal offsets between solar-dominated, thermal-dominated, and hydro-dominated grids (Section V).
2) CoordDR: a MILP formulation that jointly optimizes job dispatch (migrate / DVFS / pause / nothing) subject to

curtailment, SLA, latency, and capacity constraints (Section III).
3) A clean ablation methodology: CoordDR vs. an identical MILP with migration disabled, eliminating optimizer confounds (Section IV).
4) An evaluation across 12,840 stress-event hours showing 87–96% QoS cost reduction from adding geographic migration (Section V).

II. Related Work

A. Datacenter Demand Response

Emerald AI [4] is the closest prior system: a signal-driven orchestrator that dispatches DVFS power capping and job pausing within a single region in response to real-time DR signals. The Phoenix field trial achieved approximately 25% power reduction with zero SLA violations across 212 jobs. However, the demonstration was conducted at a single site over a short evaluation window, and Flex 0 workloads were excluded from all control actions. Our work directly extends Emerald by adding geographic migration as a control knob that targets precisely this excluded workload class.

Norris *et al.* [7] estimate that nearly 100 GW of flexible capacity could be unlocked if large consumers provide periodic curtailment. Industry analyses by Camus Energy, encoord, and Princeton [8] demonstrate that internal flexibility dispatch is needed only 40–70 hours annually for faster interconnection, but assume on-site generation or batteries rather than geographic migration. The Electric Power Research Institute's DCFlex initiative [9] partners with over 40 organizations to test demand-response and workload shifting strategies for datacenters, though cross-ISO geographic migration is not yet in scope.

B. Spatiotemporal Workload Management

The theoretical foundations of geographic load balancing were established by Liu *et al.* [10], who showed that routing workloads across geographically distributed datacenters can reduce electricity cost by exploiting regional price and renewable generation differences.

Google's Carbon-Intelligent Compute System [11] shifts flexible batch workloads within 24-hour windows using day-ahead forecasts, demonstrating fleet-level spatial coordination. RMI's analysis of datacenter flexibility [12] explicitly identifies spatial flexibility—moving delay-tolerant workloads across regions in response to grid stress—as a near-term capability, noting that the technology could be deployment-ready within 1–2 years.

CarbonClipper [13] provides a formal SOAD framework for spatiotemporal workload management with competitive ratio guarantees. However, these systems optimize for carbon intensity rather than grid load stress, and these signals can diverge: Sukprasert *et al.* [14] show that carbon-aware temporal shifting provides diminishing returns and that carbon intensity varies slowly relative to high-variance, episodic grid stress indicators. Hall *et al.* [15] propose distributionally robust optimization for fleet-wide coordination, though their DRO ambiguity sets may not hold during extreme tail events. Our work uses grid load as the primary signal and formulates the dispatch as a MILP rather than an online competitive algorithm. Ecovisor [16] virtualizes energy at the application level for per-job carbon accounting and control, but operates within a single site and does not consider cross-region migration.

C. Geo-Distributed ML Training

MAST [17] schedules ML training across geo-distributed datacenters for utilization optimization. Parcae [18] handles proactive migration of training jobs on preemptible instances. Both treat the destination as structurally underloaded—migration decisions are driven by cluster utilization, not grid state. CoordDR differs fundamentally: destination availability is dynamically determined by real-time grid headroom, which varies hourly with regional demand.

D. Live Migration Systems

The feasibility of geographic migration rests on recent systems advances. TrainMover [5] demonstrates 27–36 second live migration for 39.1B-parameter training jobs with zero memory overhead. ServerlessLLM [6] achieves 2–10 second inference pod re-warm via GPU snapshotting. We use these measured overheads directly as migration QoS penalties ($q^{\text{mig}}_{\text{train}} = 0.02$, $q^{\text{mig}}_{\text{infer}} = 0.01$), grounding our cost model in empirical data.

III. Methodology

A. System Architecture

We extend Emerald's single-site temporal framework to multi-region spatiotemporal coordination. Emerald's architecture is signal-driven: a grid operator issues a DR obligation, the Emerald Simulator evaluates strategies across three control knobs (DVFS, job pausing, resource reallocation), and the Conductor executes the optimal policy [4].

Our *Spatiotemporal Coordinator* retains the signal-driven model and adds geographic workload migration as a fourth control knob. When a DR signal arrives at a stressed region, the coordinator solves a MILP that jointly evaluates all available actions—DVFS, pausing, and migration—across the current workload. Headroom at non-stressed regions is computed from concurrent real load data, determining which migration destinations are feasible.

In our simulation, this is implemented as a per-hour dispatch: at each stress hour, the coordinator observes the current fleet state, computes headroom from real grid data at all regions, solves the MILP, and applies the optimal assignment. A production deployment would additionally require a real-time status monitor polling grid conditions and a return-migration scheduler to batch post-event job returns—these components are outside the scope of the current simulation.

B. Workload Model

Each job j in the datacenter fleet is characterized by the following attributes:

- **Job type**: training or inference.

- **Flex tier** $f_j \in \{0,1,2,3\}$: an operator-assigned flexibility class that determines the maximum allowable QoS degradation. This is a business decision, not a workload property—the same inference job may be Flex 0 (latency-critical) or Flex 3 (batch-tolerant) depending on its SLA contract.
- **Power consumption** P_j (MW): the job's current draw.
- **Home region** r_0: the region where the job is currently running.
- **Latency budget** L_j: the maximum acceptable inter-region network latency for migration. Set to 50 ms for inference and 500 ms for training, based on MLPerf benchmarks [4].

Flex tiers define the maximum QoS degradation bound φ_{f_j}: $\varphi_0 = 0$ (no degradation), $\varphi_1 = 0.05$, $\varphi_2 = 0.15$, $\varphi_3 = 0.30$. The objective weight $w_j = 1/(f_j + 1)$ makes Flex 0 jobs $4\times$ costlier to degrade than Flex 3 jobs.

We replicate the four workload ensembles from Emerald [4]: E1 (80% training / 20% inference, most Flex 0), E2 (50/50, high Flex 0), E3 (50/50, no Flex 0), E4 (90/10, moderate Flex 0).

C. MILP Formulation

1) Decision Variables: For each job j, we define a binary decision variable $z_{j,a} \in \{0,1\}$ for each action a in job j's action catalogue $\mathcal{A}_j$. Setting $z_{j,a} = 1$ assigns action a to job j.

2) Action Catalogue: The set of available actions $\mathcal{A}_j$ depends on job j's flex tier and latency budget:

- **Nothing** ($a = \texttt{noop}$): No action taken. QoS degradation $q_a = 0$; power curtailed $\delta_a = 0$.
- **DVFS at level** v ($a = \texttt{dvfs\_}v$, for $v \in \{0.9, 0.8, 0.7, 0.6\}$): Reduces GPU clock frequency to fraction v of nominal. QoS degradation $q_a = 1 - v$; power curtailed $\delta_a = P_j \cdot (1 - v)$, consistent with empirical LLM power profiling [19]. Available only if $q_a \le \varphi_{f_j}$—therefore Flex 0 ($\varphi = 0$) and Flex 1 ($\varphi = 0.05$) jobs have no feasible DVFS actions, since the mildest level imposes $q = 0.10$.
- **Pause** ($a = \texttt{pause}$): Checkpoint and suspend the job entirely. QoS degradation $q_a = 1$ (complete halt); power curtailed $\delta_a = P_j$. Available for all non-Flex 0 jobs as a *DR emergency override*—this action bypasses the per-job SLA constraint because grid compliance is contractually mandated.
- **Migrate to region** r' ($a = \texttt{migrate}_{r'}$): Live migration of the job to a non-stressed region. QoS degradation $q_a = q_j^{\text{mig}}$ (0.02 for training, 0.01 for inference, reflecting measured checkpoint/re-warm overhead); power curtailed $\delta_a = P_j$ (full power removed from stressed region). Available only if inter-region latency $\lambda_{r_0 \to r'} \le L_j$.

3) Objective: Minimize total weighted QoS degradation across all jobs:

$$\min_{z} \quad \sum_j w_j \sum_{a \in \mathcal{A}_j} q_a \, z_{j,a} \tag{1}$$

4) Constraints: **Mutual exclusivity**: each job receives exactly one action.

$$\sum_{a \in \mathcal{A}_j} z_{j,a} = 1 \quad \forall j \tag{2}$$

Curtailment compliance: total power freed meets the DR target Δ_r (a fraction of fleet power at the stressed region r).

$$\sum_j \sum_{a \in \mathcal{A}_j} \delta_a(P_j) \, z_{j,a} \ge \Delta_r \tag{3}$$

SLA compliance: no job exceeds its flex tier degradation bound, except pause actions which are exempt as a DR emergency override.

$$\sum_{\substack{a \in \mathcal{A}_j \\ a \ne \texttt{pause}}} q_a \, z_{j,a} \le \varphi_{f_j} \quad \forall j \tag{4}$$

Destination capacity: total migrated load into any region r' cannot exceed its available headroom $H_{r'}$.

$$\sum_j P_j \, z_{j,\texttt{migrate}_{r'}} \le H_{r'} \quad \forall r' \ne r_0 \tag{5}$$

where $H_{r'} = \max(0, \tau_{r'} - L_{r',t})$, with $\tau_{r'}$ the seasonal P90 load threshold and $L_{r',t}$ the current load at region r'.

The MILP is solved with the HiGHS open-source solver [20] via `scipy.optimize.milp` [21]. For 20–200 job instances, optimal solutions are found in < 50 ms, well within any practical DR dispatch interval.

D. Migration Cost Model

The migration QoS penalties are grounded in two recently published systems measurements. TrainMover [5] reports 27–36 seconds of live migration downtime for a GPT-39.1B-parameter model. Relative to the optimal checkpoint interval of $\sim$1,800 seconds identified in [22], this downtime translates to approximately 2% throughput loss over the checkpoint cycle ($q_{\text{train}}^{\text{mig}} = 0.02$). ServerlessLLM [6] achieves 2–10 second GPU-snapshot re-warm for inference pods; relative to a typical 500-second user session horizon, this is approximately 1% disruption ($q_{\text{infer}}^{\text{mig}} = 0.01$). Critically, migration transfers model state only: hyperscaler training pipelines replicate datasets across regions via object stores (e.g., S3, GCS), so the migrated job resumes from its latest checkpoint against a local data replica.

These penalties are one to two orders of magnitude smaller than even the mildest DVFS action ($q_{\texttt{dvfs\_0.9}} = 0.10$), so the MILP strongly prefers migration when headroom is available. However, because $q^{\text{mig}} > 0$, the solver migrates only as many jobs as needed to meet the curtailment target—avoiding unnecessary migrations that would accumulate QoS cost without contributing to compliance.

IV. Experimental Design

A. Region Selection

The three regions were chosen based on two criteria: (1) all major hyperscalers operate production datacenters in each, connected by private fiber; and (2) the regions have structurally different generation portfolios.

PJM covers 13 states including Northern Virginia—the world's largest datacenter market (AWS `us-east-1`, Google `us-east4`, Azure East US). PJM is thermal-dominated (gas, coal, nuclear), with stress driven by summer cooling and winter heating.

CAISO covers California (AWS `us-west-1`, Google `us-west2`, Azure West US). CAISO is solar-dominated (>50% renewable), exhibiting the "duck curve": a sharp evening stress peak as solar drops while demand remains high.

BPA covers the Pacific Northwest (AWS `us-west-2`, Google `us-west1`)—some of the largest hyperscale deployments globally. BPA is $\sim 90\%$ hydroelectric, with stress driven by water management and winter heating rather than temperature-driven cooling.

The inter-region latency matrix (measured on private networks) is: CAISO–PJM $= 70$ ms, CAISO–BPA $= 20$ ms, PJM–BPA $= 62$ ms. This implies that inference workloads (50 ms budget) can only migrate between CAISO and BPA, not cross-continent to PJM. Training workloads (500 ms budget) can migrate between any pair.

B. Data Collection and Preprocessing

We collect five years (January 2020 – December 2024) of hourly electricity demand: CAISO via the gridstatus library [23] (43,848 hours, peak 52.9 GW); PJM via PJM API v1 (43,849 hours, peak 165.5 GW); BPA via EIA API v2 (43,849 hours, peak 12.1 GW). All timestamps are aligned to UTC. Missing hours ($< 0.1\%$) are filled by linear interpolation.

C. Stress Identification

For each region r and calendar month m, we compute the load P90 threshold $\tau_{r,m}$ over all five years. An hour is marked as stressed ($s_{r,t} = 1$) if $L_{r,t} > \tau_{r,m}$ and the stress persists for ≥ 2 consecutive hours [4]. Using seasonal P90 rather than a global threshold prevents summer from dominating the classification.

D. Complementarity Metrics

We define three metrics to quantify the feasibility of spatial migration, computed as historical frequencies over the five-year dataset (not real-time predictions):

Pairwise stress correlation $\rho_{r,r'}$: Pearson correlation on binary stress indicators. Target: $\rho < 0.3$.

Simultaneous stress fraction σ_k: fraction of hours with $\geq k$ regions simultaneously stressed.

Fleet availability A_r: for each of region r's stress hours in the historical record, the fraction where ≥ 1 other region was *not* stressed at the same time. This is a retrospective empirical frequency—it quantifies how often a migration target *existed* in historical data, not a real-time prediction. Target: $A > 95\%$.

E. Simulation Setup

For each region designated as stressed, we replay all its historical stress events: 957 CAISO events (4,244 hours), 701 PJM events (4,314 hours), 760 BPA events (4,282 hours)—12,840 hours total. At each hour, a synthetic workload ensemble is generated and headroom at non-stressed regions is computed from real concurrent load data. Curtailment targets are swept from 5% to 30% of fleet power.

F. Ablation Design

The experimental design isolates the spatial contribution through a clean ablation:

- **No Coordination**: all jobs do nothing. Lower bound.
- **Temporal-Only**: the *same* MILP, same HiGHS solver, same objective and constraints—but with migration actions removed from the action catalogue. Any QoS gap vs. CoordDR is purely attributable to migration availability.
- **CoordDR**: the full MILP with all actions including migration.

On hours where no migration is possible (simultaneous stress at all regions), CoordDR and Temporal-Only solve an identical problem and produce identical solutions—confirming zero optimizer confound.

V. Results

A. Grid Stress Complementarity

Table I summarizes stress identification results. All regions exhibit ~ 9.7–9.8% stress hour rates. The critical finding is the temporal offset: CAISO stress concentrates at 00:00–04:00 UTC (evening Pacific, post-solar), PJM at 18:00–22:00 UTC (afternoon Eastern, peak thermal demand)—a 6-hour separation. BPA peaks at 20:00–00:00 UTC, decorrelated from both.

TABLE I
Stress event summary, 2020–2024.

Region	Stress hours	Events	Peak UTC hour
CAISO	4,244 (9.7%)	957	00:00–04:00
PJM	4,314 (9.8%)	701	18:00–22:00
BPA	4,282 (9.8%)	760	20:00–00:00

Fig. 1 shows the diurnal stress profiles. When CAISO stress exceeds 40% frequency (01:00–03:00 UTC), PJM stress is near zero, and vice versa.

The complementarity metrics confirm the thesis: CAISO–PJM $\rho = 0.016$, CAISO–BPA $\rho = 0.201$, PJM–BPA $\rho = 0.088$—all below 0.30. Simultaneous stress: $\sigma_2 = 4.76\%$, $\sigma_3 = 0.41\%$. Fleet availability: $A_{\text{CAISO}} = 96.2\%$, $A_{\text{PJM}} = 95.8\%$, $A_{\text{BPA}} = 96.1\%$—all above 95%.

This complementarity is structural: it arises from different generation portfolios (solar vs. thermal vs. hydro) creating distinct demand-supply dynamics, not merely from timezone differences.

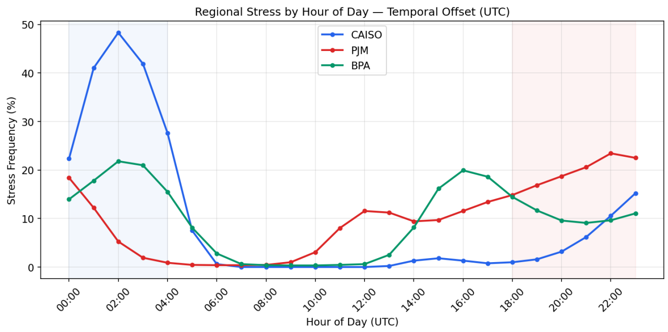

Fig. 1. Hourly stress frequency by region (UTC). The 6-hour CAISO–PJM offset is the primary driver of complementarity.

B. QoS Cost Comparison

Both CoordDR and Temporal-Only achieve the requested curtailment target across all depths (5–30%) and regions. Since both strategies use the same MILP solver and have sufficient action capacity, curtailment tracking is effectively identical—the comparison is not *how much* is curtailed, but the QoS cost of doing so.

Fig. 2 shows the central result. Temporal-Only QoS cost grows linearly from $\sim$1.75 at 5% to $\sim$12.85 at 30%—each additional 5% requires pausing or throttling more jobs. CoordDR stays near zero, rising from $\sim$0.12 to $\sim$1.0, because additional curtailment is achieved via migration at $q = 0.01$–0.02 per job.

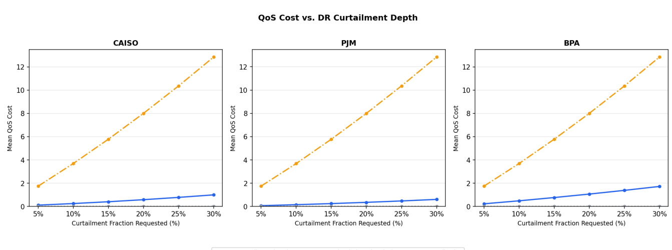

Fig. 2. QoS cost vs. curtailment depth. Temporal-Only grows linearly (more jobs paused/throttled); CoordDR remains near zero (migration at 1–2% overhead).

Table II shows the QoS reduction is consistent across depths within each region.

TABLE II
QoS COST REDUCTION FROM ADDING SPATIAL MIGRATION (E1).

Region	QoS Reduction	Mean
PJM	95.3–95.9%	95.6%
CAISO	92.2–93.3%	92.8%
BPA	86.6–86.8%	86.7%

Regional variation. PJM benefits most because its stress peaks are well-separated from both CAISO and BPA, providing abundant headroom. BPA's lower 87% reduction reflects an asymmetry: when BPA is stressed, CAISO and PJM offer enormous headroom (>10,000 MW), but the converse is not true. When CAISO or PJM is stressed and needs to send load *to* BPA, BPA's limited capacity ($\sim$12 GW peak vs. CAISO's 52.9 GW and PJM's 165.5 GW) constrains absorption. This forces CoordDR to fall back to local actions at BPA more often than at the other regions.

C. Migration as Substitution

At 10% curtailment on CAISO, CoordDR migrates 14.1 jobs/hour on average while Temporal-Only pauses 14.2 jobs/hour—nearly identical counts. The same jobs that Temporal-Only must halt ($q = 1.0$) are instead relocated ($q \approx 0.01$–0.02). At lower flex tiers, the substitution may displace DVFS rather than pause: Flex 2 jobs that Temporal-Only throttles at $q = 0.10$ via DVFS are migrated at $q = 0.02$, still yielding a 5× reduction.

On the 4.9% of hours where simultaneous stress eliminates all migration targets, both strategies produce *identical* solutions—confirming the ablation is clean.

D. Ensemble Sensitivity

Fig. 3 shows CoordDR performance across ensembles E1–E4. Results are stable. QoS cost is driven by the Flex 0 fraction: E3 (no Flex 0) achieves the lowest cost, confirming that Flex 0's constraint (migration-only) is the binding factor.

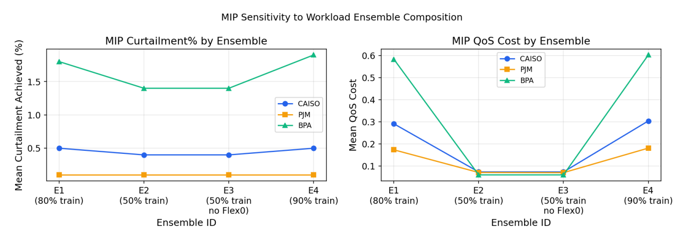

Fig. 3. CoordDR curtailment and QoS cost across ensembles E1–E4.

E. Event Case Study

Fig. 4 shows a representative CAISO event from September 6, 2022 (California heat wave) at 10% curtailment. CoordDR maintains 100% performance across all flex tiers including Flex 0 inference, while Temporal-Only degrades Flex 2 and 3 training jobs.

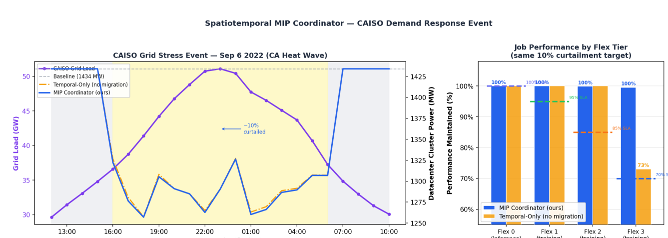

Fig. 4. CAISO stress event (Sep 6, 2022). Left: grid load and datacenter power. Right: per-tier performance.

VI. Discussion and Conclusion

A. Potential Impact

The results show that geographic migration reduces QoS cost by 87–96% at identical curtailment. This has three practical implications. *Accelerated interconnection*: a datacenter that curtails 20–30% via migration with near-zero service impact is a more attractive flexibility resource for utilities [8]. *Performance preservation*: migration maintains inference throughput and latency, making DR participation feasible for latency-critical services. *Deeper curtailment*: migration decouples curtailment depth from QoS cost, making 20–30% reductions practical.

B. Limitations

Stateless simulation. The MILP is solved independently per hour with fresh workload ensembles; no job state persists across hours. A multi-period formulation with persistent jobs would better model return migration and the no-snap-back constraint.

Synthetic workloads. Ensembles follow Emerald's compositions but use sampled power values ($\sim$8 MW training, $\sim$4 MW inference), not production traces.

Fixed migration cost. A single penalty per job type (2% / 1%) is used. Real migration cost depends on model size, checkpoint size, and network bandwidth.

Bandwidth not modeled. At fleet scales $>$500 pods, concurrent migrations could saturate WAN links.

Inference latency constraint. Flex 0 inference (50 ms budget) cannot migrate CAISO$\leftrightarrow$PJM (70 ms). Spatial flexibility for inference is limited to geographically proximate regions.

C. Threats to Validity

Internal. The ablation eliminates optimizer confounds; identical solutions on simultaneous-stress hours confirm soundness. The DR emergency override (pause bypasses SLA) inflates Temporal-Only's cost—we consider this realistic per [4], but alternative formulations would produce different numbers.

External. Results are specific to CAISO, PJM, and BPA. Regions with similar generation portfolios would show higher stress correlation and lower migration availability.

Construct. QoS cost ($\sum w_j q_j$) is a proxy; real impact depends on user-facing metrics (latency percentiles, throughput variance) that are workload-specific.

D. Future Work

Multi-period formulation with persistent jobs; real production workload traces; bandwidth-aware migration constraints; real-time grid operator API integration; extension to ERCOT, MISO, and international grids.

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